2015-Summer Paper 4 – Recent Advances

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Q1. Role of robotics in urology

Type: LAQ (25 marks)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot; "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; paediatric robotics 2002; robotic RPLND (retroperitoneal lymph node dissection) 2006.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
Standard (2000)	Three arms
S (2006) / Si (2009)	High definition, fourth arm / dual console for training
Xi (2014)	8-mm camera in any port, side docking, no stirrups
SP (2018)	All instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- Patents expired in 2019, so **Hugo, Versius** and the low-cost Indian **SSI Mantra** appeared.
- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost; cost-effective only at high volume
Tremor filtration, ergonomics	Limited haptic feedback
Less blood loss, shorter stay than open	Docking, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley, Lancet 2016** (RCT (randomised controlled trial)): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months**.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- Retzius-sparing and "hood" techniques give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.

2. RAPN (robot-assisted partial nephrectomy): versus laparoscopy, **fewer conversions**, **shorter warm ischaemia** and less eGFR (estimated glomerular filtration rate) loss. **Trifecta** = negative margin, no major complication, warm ischaemia ≤25 minutes. ICG (indocyanine green) guides selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions.
- **iROC**: **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**. Continent diversion fell (11% in iROC).

4. Other procedures

Procedure	Role of the robot
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in infants (working space about 1 L vs 6 L in adults)
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND	Template, nerve-sparing; short stay
Adrenalectomy	Equivalent to laparoscopy; suits the retroperitoneal approach
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, fistula repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

Physiology

- Steep Trendelenburg increases venous return and cardiac output, but CO_2 pneumoperitoneum causes hypercarbia and acidosis; work at 10–12 mm Hg. The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

Training

- Box trainer, **VR (virtual reality)** simulator, animal and cadaver labs, then proctoring. **FRS (Fundamentals of Robotic Surgery)** improved skills in an RCT. Five-day mini-residencies

also help. Proficiency comes after about 40 cases, but outcomes keep improving with volume.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (~22–25°)	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury (up to ~6%)	Brachial plexus (avoid shoulder braces); femoral nerve
Compartment syndrome	Long lithotomy; fasciotomy if >30 mmHg with symptoms
Bowel burns	Insulation failure, capacitive coupling; present late
Robot malfunction	About 0.4%; rarely forces conversion
Port-site hernia	Transverse incisions cut it from 5.4% to 0.4% after RARP

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

Recent advances

- Single-port transvesical surgery; **AI (artificial intelligence)** skill assessment; **telesurgery** (latency limits it).

Diagram to draw: console, vision cart and patient cart; RARP port placement.

> **Key points to remember:**
>
> - da Vinci 2000; Xi 2014; SP 2018; patents expired 2019.
> - Yaxley and Coughlin: equal function at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
> - Trendelenburg causes oedema and raised IOP; avoid shoulder braces.

Q2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation

Type: LAQ (25 marks)

Introduction

- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

Part A: Marginal kidney donor

Categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

2. KDPI (Kidney Donor Profile Index): percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of therapy	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

4. Others: **AKI (acute kidney injury)** donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en bloc); older or hypertensive living donors (KDIGO (Kidney Disease: Improving Global Outcomes) 2017 risk-based approach).

- **Absolute exclusions**: metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

Assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program)**: donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA (human leucocyte antigen) matching. Informed consent is essential.
- **Ojo (JASN 2001)**: ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI (calcineurin inhibitor) therapy.

Part B: Newer immunosuppressants

Current practice

- **Three-signal model**: signal 1 is antigen (TCR (T-cell receptor) meeting MHC (major histocompatibility complex)); signal 2 is costimulation (CD28–CD80/86); signal 3 is IL-2 receptor signalling through mTOR, driving proliferation.
- Over 90% receive induction and **tacrolimus + mycophenolate** (mycophenolate: GI (gastrointestinal) upset, teratogenic). Acute rejection is below 10%; chronic **AMR (antibody-mediated rejection)** is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Drug	Mechanism	Key points
Basiliximab	Anti-CD25 (IL-2 (interleukin-2) receptor), non-depleting	**20 mg IV (intravenous) days 0 and 4**; low risk
rATG (rabbit antithymocyte globulin)	T-cell depletion	High risk, DGF; CMV (cytomegalovirus), PTLD (post-transplant lymphoproliferative disorder)
Extended-release tacrolimus	CNI: binds FKBP12, inhibits calcineurin, so less IL-2 transcription	Once daily, better adherence; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor
Mycophenolate	Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis	Standard partner of tacrolimus
mTOR (mammalian target of rapamycin) inhibitors: sirolimus, everolimus	Bind FKBP12, block mTOR (signal 3), arresting G1 to S	Not nephrotoxic; fewer CMV/BK infections; **TRANSFORM (2018)**: everolimus with low CNI non-inferior; poor wound healing, **lymphocele**
Belatacept	**CTLA4-Ig**: binds CD80/86, blocking **CD28 costimulation (signal 2)**	FDA (Food and Drug Administration) 2011; IV 10 mg/kg induction, then **5 mg/kg 4-weekly**. **BENEFIT**: better GFR (glomerular filtration rate), more early rejection; **contraindicated if EBV (Epstein–Barr virus)-seronegative**
Imlifidase	**Cleaves IgG**	**EMA (European Medicines Agency) 2020** for crossmatch-positive, highly sensitised patients
Eculizumab	Anti-C5	Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence
Anti-IL-6 therapy	Blocks IL-6	Chronic active AMR

- **Voclosporin** is approved for lupus nephritis, **not transplantation**.

Rejection (context for drug choice)

- Types: hyperacute (preformed antibody, prevented by crossmatch), acute, subclinical, chronic. **Banff** classifies TCMR (T-cell-mediated rejection) and AMR.
- **TCMR**: pulse steroids, rATG for Banff II–III. **AMR**: steroids, IVIG (intravenous immunoglobulin), plasmapheresis, rituximab.
- **Monitoring**: donor-specific antibody, biopsy, **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%), gene-expression panels.

Recent advances

- Normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

Diagram to draw: three-signal T-cell activation: CNI after signal 1, belatacept at signal 2, basiliximab and mTOR inhibitors at signal 3, mycophenolate at DNA synthesis.

Key points to remember:

- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.
- Belatacept: better GFR, contraindicated if EBV-negative; imlifidase for highly sensitised patients.

Q3a. Etiology and management of hypospadias fistula

Type: SAQ (10 marks)

Definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak below the meatus: the **commonest complication**, just under 10%. After TIP (tubularised incised plate): 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%.

Aetiology

- **Distal obstruction** (meatal stenosis, stricture) raises voiding pressure: the **key correctable cause**.
- Ischaemia, oedema, infection, haematoma, tension.
- Technical: overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- Patient: proximal meatus, glans <14 mm, redo surgery. A long rigid neourethra raises resistance (Poiseuille's law).

Prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

Evaluation

- Split stream, usually within 2 years. Under anaesthesia: **calibration**, **urethroscopy**, and a **stress test** (saline or methylene blue) for hidden tracts.

Management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

Fistula	Repair
Small, shaft	Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier**
Large or thin tissue	Trap-door or island skin flap plus second layer
Coronal (glans dehiscence)	Redo urethroplasty and glans
With diverticulum	Excise or reuse redundant wall as flap

- Recurrence about **20%**. Catheter: none for simple, 7–14 days for complex repairs.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

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UCF → wait 6–12 months → calibrate, urethroscopy, stress test
 ├─ distal obstruction → correct at same sitting
 ├─ small → excise + layered closure + barrier flap
 └─ large/coronal → flap repair or redo urethroplasty
```

Diagram to draw: neourethra, dartos barrier and skin with offset suture lines.

Key points to remember:

- Commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

Q3b. Silodosin

Type: SAQ (10 marks)

Definition and pharmacology

- The **most α1A-selective** blocker (well over 100-fold vs α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension**. Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Flow improves in 2–6 hours, symptoms in 3–4 days. CYP3A4 metabolism.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**
CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers, PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Efficacy

- **Chapple 2011**: **non-inferior to tamsulosin 0.4 mg**, not better. Works in severe LUTS.
- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers.
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.

Adverse effects

- **Anejaculation**: over a quarter of men in trials (14% vs 2% vs tamsulosin), yet few stop; responders do better.
- No excess hypotension in the elderly or with antihypertensives. Falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome)**: reported; warn before cataract surgery.

Other uses

- **MET (medical expulsive therapy)**, distal stones 5–10 mm: faster expulsion than tamsulosin (meta-analyses); **SUSPEND (2015)** found no overall tamsulosin benefit.
- **Also know:** before ureteroscopy (easier access); CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome) of recent onset; with propiverine for storage symptoms.

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Silodosin	α1A (highest)	8 mg OD	Lowest	**Highest**
Tamsulosin	α1A, α1D > α1B	0.4 mg OD	Low	Moderate
Alfuzosin	Clinically uroselective	10 mg OD	Low	Lowest
Naftopidil	α1D	25–75 mg	Some	Less
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low

Diagram to draw: α1A prostate and vas, α1B vessels, α1D bladder.

Key points to remember:

- 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
- Equal to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect; least hypotension.

Q3c. Sacral neuromodulation

Type: SAQ (10 marks)

See 2015-Summer Paper 3, Q3a (identical question). Recent advances to add:

- **Rechargeable** stimulators (FDA (Food and Drug Administration) 2019); non-rechargeable devices lasting ≥10 years (2021).
- **Whole-body MRI (magnetic resonance imaging)-conditional** at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin).

Q3d. Urethral stents

Type: SAQ (10 marks)

Definition

- An endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.

Classification

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Not epithelialised, removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate).
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: once an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed and softens when cooled for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Removable bulbar stents succeed only in small series. UroLume worked early but failed long term from ingrowth, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no stent if urethroplasty is considered**; stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon: **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind**: nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter.

Key points to remember:

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Q3e. Studer Neobladder

Type: SAQ (10 marks)

Principles

- **Orthotopic low-pressure ileal reservoir** (**Studer, 1989**).
- **Detubularisation** stops coordinated contractions; a **sphere** (Laplace) gives most volume for surface area.
- **Isoperistaltic afferent limb** limits reflux and accepts short ureters. Continence depends on the **rhabdosphincter**.

Technique

1. Isolate **54–56 cm of ileum**, ending **15–25 cm** above the ICV (ileocaecal valve); restore bowel continuity.
2. Open the distal **40–44 cm** on the antimesenteric border; sew the two limbs into a **U** (back wall).
3. Leave the proximal **10–15 cm** tubular as the **afferent limb**; close its end.
4. **End-to-side refluxing ureteroileal anastomoses** to the afferent limb, over stents.
5. Fold the U across into a **sphere**; join its most dependent point to the urethra.

- Capacity rises from <200 to **500–600 mL** in a year.

Selection

- **Absolute contraindications**: tumour at the urethral margin, urethral stricture, severe renal or liver failure, inability to self-catheterise.
- eGFR (estimated glomerular filtration rate) **>35–40 mL/min**. Prostatic urethral tumour raises urethral recurrence risk.

Outcomes

- **Day continence 80–90%**; **night incontinence 20–50%** (no reflex, nocturnal diuresis). Self-catheterisation **10–20% of men, ~30% of women**: men leak, women retain.
- Reflux on cystogram, yet kidneys preserved in ~95%; a USC (University of Southern California) randomised trial found the antireflux T pouch no better.

Complications

- **Early**: leak, ileus, infection, mucus retention. **Late**: **hyperchloraemic acidosis**, **B12 and bile-salt loss**, ureteroileal stricture (<3%), stones, retention, hernia, pyelonephritis.

Care and follow-up

- Bicarbonate, mucus irrigation, cystogram before catheter removal. **Voiding training**: sit, relax the pelvic floor, strain; timed voiding every 2 hours, extended to 3–4 hours; **night alarm**. Annual B12, metabolic panel, upper-tract imaging.
- **Also know:** nerve and uterine sparing; intracorporeal robotic Studer.

	Studer	Hautmann
Ileum	54–56 cm, U + afferent limb	~70 cm folded in a W
Initial capacity	Smaller	Larger: earlier continence
Late	—	More retention, reabsorption

Diagram to draw: ileal segment marked, then spherical pouch with afferent limb.

Key points to remember:

- 54–56 cm ileum: 40–44 cm sphere plus 10–15 cm afferent limb.
- Day continence 80–90%, night incontinence 20–50%.
- Need eGFR >35–40 and ability to self-catheterise.

Q3f. Cryotherapy for carcinoma prostate

Type: SAQ (10 marks)

Principle

- **Transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- **Joule–Thomson effect**: **argon** freezes, **helium** thaws. Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

Cell death

- **Intra- and extracellular ice**, recrystallisation, **vascular stasis and ischaemia**, apoptosis, immune effects.
- **−40 °C**, fast freeze, slow thaw, **two cycles**; the ice-ball edge is sublethal.

Indications

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.
2. **Primary whole-gland**: men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland**: MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.

Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT (randomised controlled trial)): **non-inferior** to external beam radiotherapy; worse sexual function.

Complications

	Primary / focal	Salvage
ED (erectile dysfunction)	**Almost always** (whole gland)	Very common
Incontinence	Rare	Single digit
Rectourethral fistula	≤0.8% (focal)	1–3.4%
Retention	Commonest after focal	~4%

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD (Cryo On-Line Data) registry).

	Cryotherapy	HIFU	IRE (irreversible electroporation)
Energy	Freezing	Focused ultrasound heat (≥55 °C)	Electric pulses, non-thermal
Route	Transperineal	Transrectal	Transperineal
Best for	Anterior lesions	Posterior lesions	Lesions near nerves, vessels

Guidelines and advances

- **EAU (European Association of Urology)**: primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

Diagram to draw: prostate with ice ball, −40 °C isotherm inside 0 °C edge.

Key points to remember:

- Argon freezes, helium thaws; −40 °C, two cycles, urethral warmer.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

Rapid Revision Sheet – 2015-Summer Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019.
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- Commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

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- 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
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